

Coupling crop growth models and machine learning for scalable winter wheat yield estimation across major wheat regions in China

Jiyuan Xie^{a,b}, Dongyan Zhang^{a,b,*}, Ning Jin^c, Tao Cheng^{a,b}, Gang Zhao^d,
Dong Han^{a,b}, Zhen Niu^{a,b}, Weifeng Li^{a,b}

^a College of Mechanical and Electronic Engineering, Northwest A&F University, Yangling, Shaanxi 712100, PR China

^b Shaanxi Key Laboratory of Agricultural Information Perception and Intelligent Service, Northwest A&F University, Yangling, Shaanxi 712100, PR China

^c Department of Resources and Environmental Engineering, Shanxi Institute of Energy, Jinzhong 030600, PR China

^d State Key Laboratory of Soil Erosion and Dryland Farming on the Loess Plateau, Northwest A&F University, Yangling, Shaanxi 712100, PR China

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ABSTRACT

The temperate continental climate, with its combination of ample sunlight and relatively low humidity during key growth stages, provides favorable conditions for wheat cultivation, reducing disease risks and supporting efficient grain filling. However, spatial heterogeneity in soil properties, crop varieties, and agronomic practices complicates large-scale yield prediction. This study focuses on China, the world's largest wheat producer, to develop an adaptable method for estimating yields in major wheat-growing regions. We coupled the WOFOST model with HYDRUS and CASA to simulate crop growth, soil moisture, and carbon cycling, assimilated remote sensing data for model calibration, and trained a machine learning model using multi-modal data to predict regional-scale winter wheat yields. The model, combining a Bagging Regressor with total above ground production (TAGP), gross primary productivity (GPP), and evapotranspiration (ET), achieved optimal performance ($R = 0.83$, $RRMSE = 0.12$) and reduced prediction errors compared to Global-WheatYield4km, with an average error of 1.92%. The model also reliably predicted yields across varying conditions, with discrepancies under 1,000 kg/ha in most counties. This integrated approach enhances yield prediction stability and precision, offering a solid foundation for regional-scale agricultural management.

1. Introduction

In the context of climate change and population growth, the accurate and efficient prediction of crop yield within the region is of paramount importance for the sustainable production of food and the security of the global food supply (Rezaei et al., 2023). Early yield forecasts provide significant advantages, such as informing agricultural policy, optimizing resource allocation, and enhancing economic efficiency. Wheat, a staple crop in China, thrives in regions marked by low annual precipitation and considerable seasonal temperature fluctuations. These conditions underpin consistent food production, making these areas crucial not only to China's food security but also to global wheat supply chains, thereby contributing to the stability of international food markets (Zhang et al., 2019). However, the spatial heterogeneity of soil properties, along with variability in management practices and meteorological conditions, presents substantial challenges to accurate large-scale yield prediction.

Research on crop yield prediction has consistently been a focal area in the agricultural domain, primarily encompassing two methodological categories: field-based survey estimation approaches and remote sensing-based estimation techniques. Empirical models, while straightforward, are limited by human experience and field-based investigations, making them difficult to apply in complex and variable environments (Lobell and Burke, 2010). In contrast, remote sensing-based models use multi-source, high-resolution data from satellite platforms to monitor crop growth and estimate yield. Currently, remote sensing-based methods for crop yield estimation are primarily categorized into two methodological approaches: Data Assimilation (DA), which integrates remote sensing data into crop models, and Remote Sensing Indices Systems (RSIS), which estimate yield based on remote sensing-derived indices. The DA approach has gained popularity for large-scale yield estimation, as it combines crop growth models with remote sensing data to improve accuracy (Manivasagam and

* Corresponding author.

E-mail address: zhangdy@nwfau.edu.cn (D. Zhang).

¹ Present/permanent address: College of Mechanical and Electronic Engineering, Northwest A&F University, Yangling, Shaanxi 712100, PR China

Rozenstein, 2020). For example, the integration of Sentinel-2's leaf area index (LAI) into crop models has shown strong predictive performance in estimating field yield variability (Gasó et al., 2021). RSIS models often use regression techniques to predict yields from remote sensing indicators, with machine learning algorithms enhancing their capacity to handle nonlinear relationships and large datasets (Cao et al., 2021).

Despite their advantages, both methods have limitations. DA models are highly dependent on the quality of input data, which can limit predictive accuracy, especially under complex conditions. While RSIS models are more efficient, they tend to rely on simplified indices that fail to capture the full complexity of crop growth dynamics. As a result, there is increasing interest in combining these two approaches to leverage their complementary strengths. For instance, integrating machine learning algorithms to adjust crop model outputs can improve yield estimates, especially under extreme weather conditions (Bai et al., 2024).

In crop growth modeling, the WOFOST model has demonstrated strong performance due to its ability to simulate crop growth with relatively minimal data input requirements. However, it has limitations in modeling soil water dynamics and the carbon cycle—both critical factors affecting yield prediction. Soil moisture plays a vital role in crop growth, yet current models struggle to simulate its dynamics accurately, thus impacting yield predictions (Huang et al., 2024). Additionally, the carbon cycle, which influences physiological processes like material deposition, is not sufficiently incorporated into most models (Fattori Jr and Marin, 2023).

While machine learning techniques have shown promise in improving the accuracy of remote sensing-based yield predictions, they still struggle to address the complexity of environmental interactions and data relationships, limiting their full potential (Yuan et al., 2020). Furthermore, within temperate continental climates, meteorological conditions, including precipitation, temperature extremes, and humidity, can vary significantly at regional and local scales, affecting soil, crop growth, and agronomic practices. These variations lead to considerable yield differences across regions with similar broad climatic conditions (Wang et al., 2017). Given these challenges, there is a growing need for a more integrated yield prediction system that combines specific climate, remote sensing, soil, and crop data across different growth stages. This system should incorporate crop simulation models and advanced machine learning techniques to enhance prediction precision and reliability.

To address these gaps, this study aims to optimize existing crop growth models and integrate remote sensing data and machine learning algorithms to develop an improved wheat yield estimation method. The specific objectives are: (1) coupling the WOFOST model with the HYDRUS and CASA models to comprehensively simulate crop growth, soil moisture dynamics, and the carbon cycle; (2) assimilating remote sensing data and calibrating the crop model for improved accuracy; (3) developing a multi-index machine learning framework for regional yield prediction; and (4) evaluating the robustness and adaptability of the model across different wheat-producing areas, accounting for variations in soil, crop growth, agronomic practices, and local meteorological conditions in temperate continental climates.

2. Materials and data

2.1. Research area and technical route

The study area encompasses 11 major winter wheat production provinces in China, comprising 1,363 county-level administrative regions. These include Henan, Shandong, Anhui, Jiangsu, Hebei, Hubei, Shanxi, Shaanxi, Sichuan, Xinjiang and Gansu (Fig. 1). The winter wheat planting area in these provinces represents over 93% of China's total winter wheat planting area, with the total winter wheat production accounting for over 96% of the country's wheat production (Yearbook, 2021). The climate types present in the study area are diverse, with a considerable range of temperate monsoon climates observed in the northeast. The southeastern part of the region exhibits a subtropical monsoon climate, with a clear distinction between the four seasons observed in the central and western regions. The climate in these areas is classified as temperate continental. The sowing of winter wheat in the study area commenced at the beginning of October and the harvest was completed before July. Fig. 2 presents the workflow of multimodal data preprocessing and model construction, illustrating the technical approach for integrating and analyzing data to model winter wheat yields effectively.

2.2. Data sources

2.2.1. Coupled crop model and input data acquisition

The HYDRUS-WOFOST-CASA coupling model requires the inputs of meteorological data, soil data, crop data, and terrain and land use data.

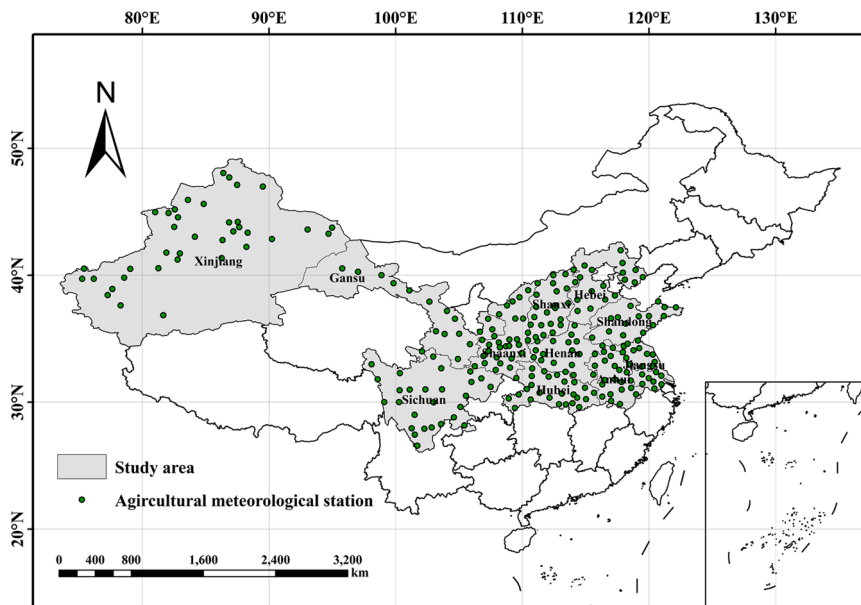


Fig. 1. Study area and locations of winter wheat agricultural meteorological stations.

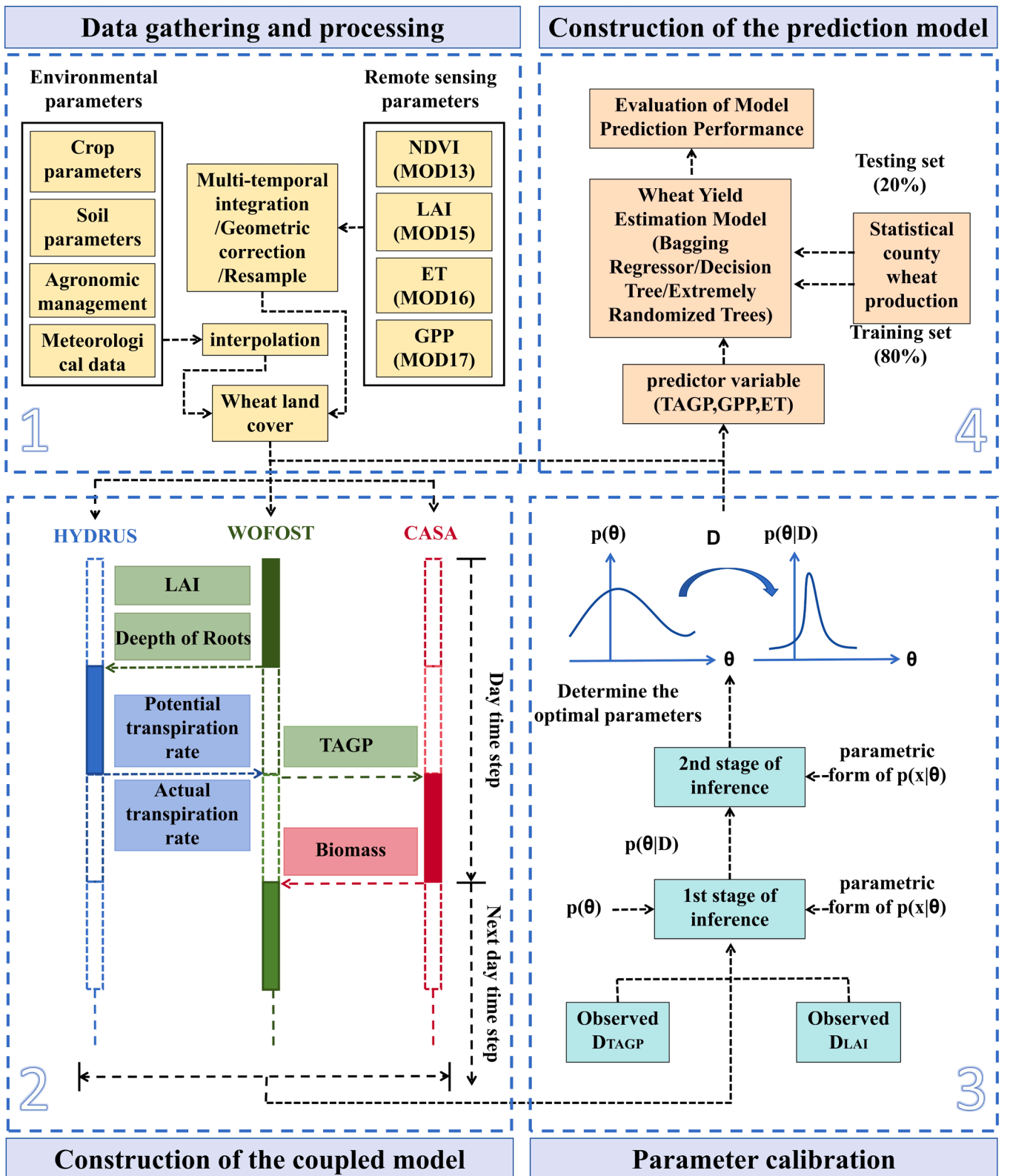


Fig. 2. Flowchart of data acquisition, processing, and research methodology.

The distribution of agricultural meteorological stations selected for analysis in this study is illustrated in Fig. 1, with data spanning the period from 2001 to 2020. The agricultural meteorological station provided data from field tests on winter wheat growth, and the specific settings for simulation according to agronomic management measures are presented in Table 1. The soil characteristics data were derived from

the 1:100,000 Chinese soil data set, which is included in the World Soil Database (HWSD). The data pertaining to the planting area of winter wheat are sourced from the National Ecological Science Data Center, with a spatial resolution of 30 m. The data has been verified through the utilisation of field survey samples and Google Earth samples. The overall accuracy of winter wheat identification is 91.17%. The meteorological

Table 1
Coupling model agronomic management parameter setting.

Management measures	detail setting
Sowing period	October 1 st , October 10 th , October 20 th , October 30 th
Sowing volume	12.5 kg, 15 kg, 17.5 kg, 20kg
Irrigation	No irrigation, winter irrigation, jointing irrigation, winter irrigation + jointing irrigation (irrigation amount is 450 m ³ /ha).
Fertilization amount	N:0 kg, 2.5 kg, 5 kg; P:0 kg, 2.5 kg, 5kg
Varieties parameters	According to the main varieties recorded in the local agricultural site, two varieties were set up, and the key parameters fluctuated by 5 % compared with the main varieties.

forcing data set for China serves as the primary meteorological driving factor for the coupled model (He et al., 2020). Furthermore, supplementary weather data were obtained from the NASA Power database (Table 2).

The maximum light use efficiency of winter wheat used in the CASA model in this study comes from (Liu ZhenZhen et al., 2017). The 30-minute continuous flux observation data used for model accuracy verification were obtained from eight representative ecological stations in ChinaFLUX.

2.2.2. Coupled crop model data acquisition

The MOD13Q1 NDVI product was selected as the remote sensing data input into the Net Primary Production (NPP) estimation model. The MODIS Reprojection Tool was employed for the purposes of projection conversion, image fusion and S-G filtering on the data. The monthly NDVI data, with a spatial resolution of 500 m, from 2001 to 2020 were employed as model input parameters.

In the process of coupling the model simulation, remote sensing data is employed for data assimilation and parameter calibration, thereby

Table 2
Meteorological driving factors required for the coupling model.

Parameter name	Defines	Unit	Data source
IRRAD	Daily global radiation	J/ m ² / day	Srad × 24 × 3.6/1000 (conversion from China Meteorological Forcing Dataset)
TMIN	Daily minimum temperature	°C	min(Temp)-273.15 (conversion from China Meteorological Forcing Dataset)
TMAX	Daily maximum temperature	°C	max(Temp)-273.15 (conversion from China Meteorological Forcing Dataset)
VAP	Mean daily vapour pressure	hPa	Shum × Pres/(0.622+Shum × 0.378)/10000 (conversion from China Meteorological Forcing Dataset)
RAIN	Precipitation (rainfall or water equivalent in case of snow or hail)	cm/ day	Prec × 24/10 (conversion from China Meteorological Forcing Dataset)
WIND	Mean daily wind speed at 2 m above ground level	m/ sec	Provided by China Meteorological Dataset, no conversion required
E0	Penman potential evaporation for a free water surface	cm/ day	Provided by NASA Power database, no conversion required
ES0	Penman potential evaporation for a bare soil surface	cm/ day	Provided by NASA Power database, no conversion required
ETO	Penman or Penman-Monteith potential evaporation for a reference crop canopy	cm/ day	Provided by NASA Power database, no conversion required

facilitating the acquisition of a more accurate high-order index-TAGP. A yield prediction model was constructed using remote sensing data, combining two key indicators, namely GPP and ET. The model was calibrated using winter wheat yield data at the county level as the target variable.

In this study, MOD15A2 LAI products were employed for data assimilation, and 8d synthetic crop ET and GPP data were extracted from MOD16A2 ET products and MOD17A2 GPP products. The county-level grain yield data were derived from the statistical yearbook for the period between 2001 and 2020. As the majority of statistical yearbooks do not record yield data directly, this study has obtained county-level yield data by dividing the total yield by the planting area (Huang et al., 2022). The resulting unit of yield data is kg/ha. Detailed information of the above data is provided in the following Table 3

3. Methodology

3.1. HYDRUS-WOFOST-CASA coupling model

The WOFOST model is a widely utilised tool for the simulation of

Table 3
Sources of data.

Data type	Time resolution	Spatial resolution	Data source
Daily weather variables	Daily (Cumulated to eight days)	0.25° (Bi-linear interpolation to 500 m)	NASA Power database (https://power.larc.nasa.gov/)
The China Meteorological forcing dataset	3-hour (Cumulated to eight days)	0.1° (Bi-linear interpolation to 500 m)	https://doi.org/10.6084/m9.figshare.11558439
Normalized Difference Vegetation Index (NDVI)	16-day (Cumulated to one month)	250m (Resolution reduced to 500 m)	NASA, MOD13Q1 NDVI product (http://ladswb.modaps.eosdis.nasa.gov/)
LAI	8-day	500m	NASA, MOD15A2 LAI product (http://ladswb.modaps.eosdis.nasa.gov/)
ET	8-day	500m	NASA, MOD16A2 ET product (https://ladswb.modaps.eosdis.nasa.gov/)
GPP	8-day	500m	NASA, MOD17A2 GPP product (https://ladswb.modaps.eosdis.nasa.gov/)
Wheat planting distribution data	Yearly	30m	National Science & Technology Infrastructure of China (http://www.nesdc.org.cn)
Record yield	Yearly	County level	China Economic Social Big Data Research Platform (https://data.cnki.net/)
Measured yield	Yearly	Point scale	National Meteorological Data Center of China (http://data.cma.cn)
Site observation data	Daily (Cumulated to eight days)	Point scale	National Meteorological Data Center of China (http://data.cma.cn)

crop growth. The WOFOST model is capable of quantitatively simulating the growth process of crops under the influence of meteorological and other environmental factors, and of effectively predicting the growth process and yield of crops. It has been employed in numerous fields across a multitude of countries and regions (Abebe et al., 2022). Nevertheless, the model exhibits certain limitations in its capacity to comprehensively simulate soil water dynamics and the carbon cycle. In particular, the WOFOST model is relatively simplistic in its treatment of soil moisture, and it is challenging to accurately simulate soil moisture changes under complex meteorological conditions (Wu et al., 2021). Furthermore, the WOFOST model is primarily concerned with crop biophysical processes, and thus has limited capacity to simulate soil moisture dynamics and the integrated carbon cycle.

In order to address these limitations, this study integrated the WOFOST model with the HYDRUS and CASA models to develop a comprehensive simulation of crop growth, soil moisture dynamics, and the carbon cycle (Fig. 3). The HYDRUS model is designed to simulate the movement of water and solutes in soil. It can provide more detailed data on soil water distribution and its changes, thereby improving the accuracy of predictions regarding the impact of soil moisture on crop growth. The HYDRUS model is founded upon the Richards equation, which forms part of the saturated-unsaturated convection-dispersion equation under a range of conditions. This enables a quantitative analysis of soil moisture status and crop water use efficiency.

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K \left(\frac{\partial h}{\partial z} + 1 \right) \right] - S \quad (1)$$

Among them, θ is the soil volumetric water content; t is time; Z is the spatial coordinate; K is soil hydraulic conductivity; h is pressure head; S is root water absorption. Soil volumetric water content $\theta(h)$ and soil hydraulic conductivity $K(h)$ were calculated by Van Genuchten formula.

$$\theta(h) = \begin{cases} \theta_r + \frac{\theta_s - \theta_r}{[1 + (ah)^n]^m}, & h < 0 \\ \theta_s, & h \geq 0 \end{cases} \quad (2)$$

$$K(h) = K_s S_e^l \left[1 - \left(1 - S_e^m \right)^m \right]^2 \quad (3)$$

Among them, θ_s is saturated water content; θ_r is retention water content; α and n are the empirical parameters that determine the shape of the water characteristic curve, where $m = 1 - \frac{1}{n}$; K_s is the saturated water content; l is the soil pore connectivity parameter, which is determined according to the default management parameters. S_e is relative saturation.

In this study, the Feddes function was selected as the water stress function, with the objective of simulating the water absorption process of wheat roots. The upper boundary condition is defined as the atmospheric boundary of integrated water covered by crops. Given that the groundwater level in the study area is 30 to 60 metres, which is considerably greater than the thickness of the observed soil layer, the lower boundary condition is set as a free drainage state. In accordance with the soil properties, the 10-meter profile was partitioned into five layers through the implementation of spatial and temporal discretization. The soil hydraulic parameters were established in accordance with the recommendations set forth in the relevant literature (Li et al., 2019), and the profile was divided into 100 units at an interval of 10 cm. In accordance with the aforementioned methodology, 101 nodes and 5 observation points were established within the soil profile. The initial time interval of the model is set to 0.01 days, the minimum time interval is set to 0.001 days, and the maximum time step is set to 1 day.

Concurrently, this study introduces the light energy utilisation model, which serves to enhance the simulation of the carbon cycle. The light energy utilisation model is based on remote sensing data to simulate the dry matter quality of crops, and this is then combined with the expression relationship, such as the harvest index, to obtain a remote sensing estimation modelling approach of crop yield per unit area. The CASA model is one of the most widely utilised models in the field. The biomass formula during the crop growth period is as follows:

$$Biomass = \sum_{t=t_0}^{t_n} NPP(x, t) = \sum_{t=t_0}^{t_n} (APAR(x, t) \times \varepsilon(x, t)) \quad (4)$$

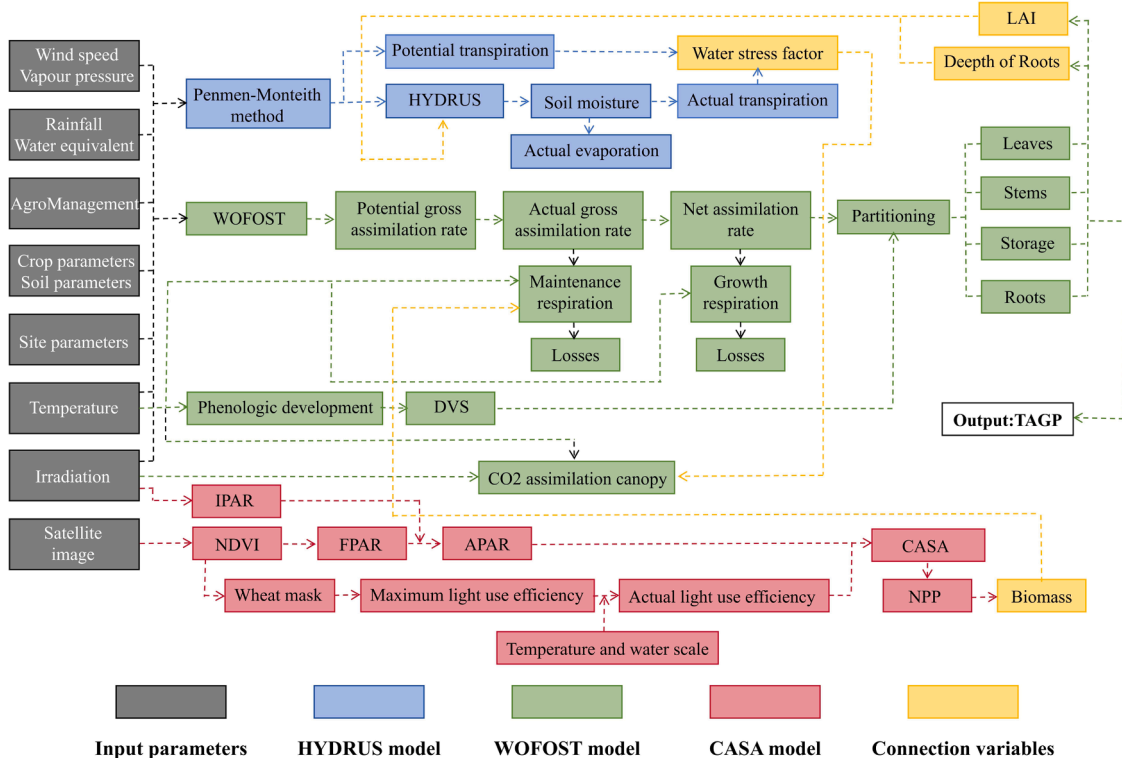


Fig. 3. HYDRUS-WOFOST-CASA coupling model structure.

$$APAR(x, t) = PAR(x, t) \times FPAR(x, t) \quad (5)$$

$$\varepsilon(x, t) = T_{e1}(x, t) \times T_{e2}(x, t) \times W_e(x, t) \times \varepsilon_{\max} \quad (6)$$

Among them, NPP is the net primary productivity during the growth period of the crop, t is the different growth periods of the crop, in monthly units, x is the spatial position of the crop growth, t_0 and t_n indicate the beginning of the growth and harvest stage of the crop, $APAR$ is photosynthetically active radiation, ε is the actual light energy utilization rate. PAR is photosynthetically active radiation, calculated according to the formula given by the World Food and Agriculture Organization (RAYMOND HUNT, 1994); $FPAR$ is the proportion of photosynthetically active radiation absorption, which is estimated by NDVI (Zhu et al., 2007); T_{e1} and T_{e2} represent the stress effect of low temperature and high temperature on light energy utilization rate. The average temperature of the month corresponding to the maximum MODIS NDVI is selected as the optimum temperature, and the calculation formula is based on the literature (Potter et al., 1993). W_e is the influence coefficient of water stress, which reflects the influence of water conditions. This module has been integrated into the soil moisture sub-model of CASA model.

Following the introduction of the CASA model, it is possible to simulate the carbon dynamics of the ecosystem with greater precision, including aspects such as plant productivity, carbon fixation and soil respiration. This enables the coupled model to provide a more comprehensive description of the absorption, storage and release of carbon during crop growth, thereby facilitating a more accurate assessment of the carbon cycle.

The combination of the WOFOST model with the HYDRUS and CASA models allows for the effective overcoming of the limitations of a single model in soil moisture dynamics and carbon cycle simulation. This study employs a two-way loosely coupled approach to model construction, as illustrated in Fig. 2. Firstly, the WOFOST model is driven by meteorological, crop, soil and field management data, with a time step size of one day. Subsequently, the parameters, including LAI and root depth, generated by the WOFOST model were introduced into the HYDRUS model to simulate the alterations in soil water balance and soil water content. The input of the WOFOST model was then updated with the potential and actual transpiration output by the HYDRUS model. The CASA model employs information on NDVI value, light energy utilisation rate and total solar radiation in wheat planting areas to simulate the net primary productivity of crops. Subsequently, the crop harvest index is constructed to facilitate the conversion of biomass and yield. However, the spatial and meteorological heterogeneity in China is considerable, and it is not feasible to achieve this process solely through a formula. Consequently, following the calculation of the crop biomass simulated by the CASA model, this study deducts the dry matter weight of the crop respiration consumption and redistribution. The dry matter distribution process of the WOFOST model is then employed in lieu of the harvest index to simulate the total aboveground biomass of the crop.

In consideration of the disparate meteorological conditions that prevail in the study area (Beck et al., 2023), two distinct simulation scenarios pertaining to the irrigation and supplementary irrigation areas have been established in the present study. In these areas, irrigation districts rely on systematic water supply from hydraulic infrastructure, whereas supplementary irrigation zones depend predominantly on natural precipitation with minimal supplemental irrigation during critical growth phases. The irrigation areas encompass Shanxi, Sichuan, Gansu, Xinjiang, Hubei, portions of Anhui, southern Jiangsu, and central and northern Shaanxi. The supplementary irrigation areas includes Henan, Shandong, Anhui, northern Jiangsu, Hebei, and southern Shaanxi. The simulation period is 20 years (2001–2020). A total of 27, 260 simulations were conducted at the county level, based on data from agricultural stations. The specific agronomic management settings are presented in Table 2.

3.2. Data assimilation and parameter calibration

At the regional scale, the assimilation and calibration of parameters are beneficial for the effective adaptation to spatial heterogeneity and the improvement of the model's predictive performance. This study proceeded to optimise the pivotal parameter that regulates the phenological development phase of crops LAI. The mean value of LAI remote sensing observations within a 500 m radius of the agricultural meteorological station is employed as the actual observation value. This is then integrated with the simulation results of the coupled model, which are subsequently assimilated using the Ensemble Kalman Filter (EnKF). This process enhances the precision of the simulated LAI parameters.

In this study, the calibration of parameters was also achieved at the level of each individual region. In light of the lack of notable alterations in the records of agricultural meteorological stations from 2001 to 2020 pertaining to winter wheat varieties, it is postulated that the parameters associated with the winter wheat crop remain unaltered on a five-year cycle. The average statistical yield of each county from 2001 to 2020 was calculated and converted into dry matter mass (calculated at 12.5% standard moisture content) based on the observation data collected every five years. The county-level yield was converted to TAGP for calibration, in conjunction with the harvest index of each province. The harvest index of each province was estimated based on the dynamic growth records of the biomass composition of the dominant winter wheat varieties in the province, as documented by the agricultural meteorological station (Xie Guanghui et al., 2011). The posterior function is maximised based on the statistical data of the maximum leaf area index and the TAGP, which is derived from the county-level yield every five years. This allows for the optimisation of the selected parameters.

Assume that the prior distribution $P(\theta)$ of parameter θ is a multi-variate normal distribution with mean vector μ_θ and covariance matrix Σ_θ :

$$P(\theta) = \frac{1}{(2\pi)^{\frac{k}{2}} |\Sigma_\theta|^{\frac{1}{2}}} e^{-\frac{(\theta - \mu_\theta)^T \Sigma_\theta^{-1} (\theta - \mu_\theta)}{2}} \quad (7)$$

Observation data D contains two parts: LAI observation data D_{LAI} and TAGP observation data D_{TAGP} . The maximum LAI observations from all agricultural meteorological stations within the study area were collated and presented in histogram form. The results demonstrate that the histogram adheres to a normal distribution, with a mean value of 6.37 and a standard deviation of 1.29. In light of the aforementioned evidence, we define the likelihood function as follows:

$$P(D|\theta) = P(D_{LAI}|\theta) \cdot P(D_{TAGP}|\theta) \\ = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_{LAI}^2}} e^{-\frac{(y_{LAI,i} - f_{LAI}(\theta))^2}{2\sigma_{LAI}^2}} \cdot \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_{TAGP}^2}} e^{-\frac{(y_{TAGP,i} - f_{TAGP}(\theta))^2}{2\sigma_{TAGP}^2}} \quad (8)$$

Among them, $y_{LAI,i}$ and $y_{TAGP,i}$ are the i observed values, $f_{LAI}(\theta)$ and $f_{TAGP}(\theta)$ are the predicted values of LAI and TAGP under parameter θ , respectively, and σ_{LAI} and σ_{TAGP} are the standard deviations of the observed data. According to the Bayesian theorem, the posterior probability $\log P(\theta|D)$ can be expressed as:

$$\log P(\theta|D) \propto \log P(D|\theta) + \log P(\theta) \\ = -\frac{n}{2} \log(2\pi\sigma_{LAI}^2) - \frac{1}{2\sigma_{LAI}^2} \sum_{i=1}^n (y_{LAI,i} - f_{LAI}(\theta))^2 \\ - \frac{n}{2} \log(2\pi\sigma_{TAGP}^2) - \frac{1}{2\sigma_{TAGP}^2} \sum_{i=1}^n (y_{TAGP,i} - f_{TAGP}(\theta))^2 \\ - \frac{1}{2} \log(2\pi)^k \left| \sum \theta \right| - \frac{1}{2} (\theta - \mu_\theta)^T \Sigma_\theta^{-1} (\theta - \mu_\theta) \quad (9)$$

The gradient descent method is used to maximize the logarithmic

posterior probability to determine the optimal parameter θ^* .

$$\theta^* = \operatorname{argmax}_{\theta} P(\theta|D) \quad (10)$$

The entire process entails the integration of prior knowledge and observational data, with the objective of optimising the selected parameters of the model through the maximisation of the posterior probability.

3.3. Wheat yield prediction model

The HYDRUS-WOFOST-CASA coupling model was employed to simulate the multi-year and multi-phenological period in the principal wheat-cultivating regions of China. The data assimilation and parameter calibration were conducted in accordance with satellite image data and statistical data. The Bagging Regressor method was employed to integrate the output parameters of the coupling model with a range of remote sensing indicators, with the objective of developing a wheat yield prediction model. The model flow chart is presented in Fig. 2, and the specific process is outlined below:

- (1) A simulation of the coupling model is presented herewith. The spatial data points of agricultural meteorological stations are constructed as triangulation networks, and the data are assigned by the nearest neighbour interpolation method to determine different simulation conditions. Subsequently, the coupling model is executed to generate a substantial number of simulation parameters, and the pertinent data from satellite images is employed for data assimilation. In the course of the simulation, the observed LAI value was employed for data assimilation, and the scale of the integrated model was set to 50, with the random seed fixed at 10000. The findings indicate that this represents an optimal balance between the reduction of the EnKF random element and the computational burden associated with the algorithm (Popov et al., 2021). The WOFOST72 WLP FD model is employed for the initialization of multiple model instances. Upon reaching the designated observation date, the state data for each model is duly collected. Subsequently, the Kalman gain matrix is calculated and the state estimation is updated through the assimilation process. Once the simulation has been completed, the parameters are calibrated using the previously constructed posterior function in order to accurately reflect the crop growth process under different meteorological conditions and obtain more accurate TAGP values.
- (2) Construct a yield prediction model. In consideration of the dynamic characteristics of wheat growth, the simulated TAGP values of the three phenological periods were selected as predictors. The model incorporates three key stages of wheat growth: the heading stage (occurring 80-64 days before harvest), the filling stage (occurring 64-40 days before harvest) and the maturity stage (occurring 40 days before harvest, up until the day of harvest). Additionally, GPP and ET were selected, and the three physiological and remote sensing indexes were employed as model inputs individually or in conjunction. Wheat yield was employed as the target variable in the construction of the prediction model, which was established using a machine learning algorithm. Three machine learning models were selected for model construction. The Decision Tree Regressor divides the data by decision rules for regression prediction, while the Extremely Randomized Trees Regressor builds a regression tree based on randomly selected features and segmentation points, which is usually faster and more accurate than traditional random forest training. The Bagging Regressor is an ensemble method that enhances accuracy and stability by training multiple models and averaging their predictions. The objective is to ascertain which machine learning method is most effective for each indicator across different prediction periods. Subsequently, the optimal

yield prediction model, which was constructed in the local engine, is employed in regional image synthesis to predict the yield of major wheat-producing regions in China.

3.4. Model evaluation

Using statistical data, the model is verified at the county scale. Three indicators are selected to quantify the performance of the model: coefficient of determination R^2 , relative root mean square error $rRMSE$ and relative error RE . The calculation formula of these indicators is:

$$R^2 = 1 - \frac{\sum_{i=1}^n (O_i - P_i)^2}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (11)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (O_i - P_i)^2} \quad (12)$$

$$rRMSE = \frac{RMSE}{\bar{O}} \quad (13)$$

$$RE = \frac{P_i - O_i}{O_i} \times 100\% \quad (14)$$

where O_i and P are observation data and prediction data, $\bar{O}$ is the average value of observation data, and n is the number of samples.

Moran's index and local Moran's index are statistical tools employed to assess spatial autocorrelation in the context of spatial statistical analysis. Moran's index is employed to quantify the global autocorrelation of a variable in space. The calculation formula is as follows:

$$I = \frac{N \sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2 \sum_j \sum_i w_{ij}} \quad (15)$$

The Local Moran Index is employed for the purpose of evaluating the spatial autocorrelation of each geographical unit. The calculation formula is as follows:

$$I = \frac{N \sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2 \sum_j \sum_i w_{ij}} \quad (16)$$

Among them, x_i and x_j are the data values of regions i and j , $\bar{x}$ is the mean value of the data, w_{ij} is the weight matrix (usually representing the spatial adjacency relationship of regions i and j), N is the number of regions, and $\sum_i$ is the sum of all adjacent regions. The positive values of the two indexes indicate that the data values of the region exhibit a tendency towards spatial aggregation in its adjacent region. A negative value indicates that the data value of the region is characterised by a spatial dispersion phenomenon in its adjacent region.

4. Results

4.1. Data correlation analysis

In order to refine the model parameter settings, principal component analysis (PCA) was performed on the meteorological data selected in this paper. Figure S1 illustrates the distribution of the input meteorological variables and the output TAGP on the first principal component (PC1) and the second principal component (PC2) of the coupled model. The results demonstrated that the cumulative explained variances of PC1 and PC2 were 54.75% and 26.35%, respectively, which collectively explained 81.10% of the data variance and effectively captured the primary variation of the data. It is evident from the figure that there are significant disparities in the distribution of meteorological data across different provinces. Consequently, it is essential to establish comparable parameters for provinces with analogous climatic characteristics in the

coupling model simulation.

The data obtained from the machine learning analysis were classified into five distinct groups based on the latitude of yield, with each group representing a specific yield range: 3000 kg/ha, 4500 kg/ha, 6000 kg/ha, 7500 kg/ha, and 9000 kg/ha (Fig. 4). The results of the scatter plot and correlation coefficient analysis indicated a significant positive correlation between yield and TAGP and GPP. The correlation coefficient between yield and TAGP is 0.83, which provides further evidence of the necessity of utilising a coupled crop model for the fine simulation of TAGP in order to predict yield. The kernel density estimation plot demonstrates the probability density distribution of each variable across different yield level groups. The presence of multiple peaks may indicate the diversity or grouping of data. For instance, the yield levels of 3000 kg/ha and 4500 kg/ha may reflect disparate planting conditions or management strategies. The bimodal phenomenon observed in the 6000 kg/ha level group may be attributed to the influence of varying regions or production systems within the study area.

4.2. Performance of coupling model

The WOFOST model and the HYDRUS-WOFOST-CASA coupling model are employed to facilitate a comparative analysis of the simulated and measured days of wheat phenology across 11 agricultural stations situated in Linying, Huyi, Wenxi, Danjiangkou, Gaoyi, Zhangqiu, Shucheng, Pidu, Jiangyan, Yuzhong and Shache, as illustrated in Fig. 5. The various subgraphs illustrate the different stages of fertility. The green dots represent the results of the coupling model, while the red dots represent the results of the WOFOST model. The dotted line represents a ratio of 1:1, indicating that the simulated value is perfectly aligned with the measured value under ideal conditions.

The HYDRUS-WOFOST-CASA coupling model of the green point (R^2 0.74-0.81, rRMSE 0.05-0.11) in terms of the fitting degree R^2 (0.79-0.87) and the error rRMSE (0.04-0.08) at each growth stage. This indicates that the coupling model has a notable enhancement in simulation accuracy.

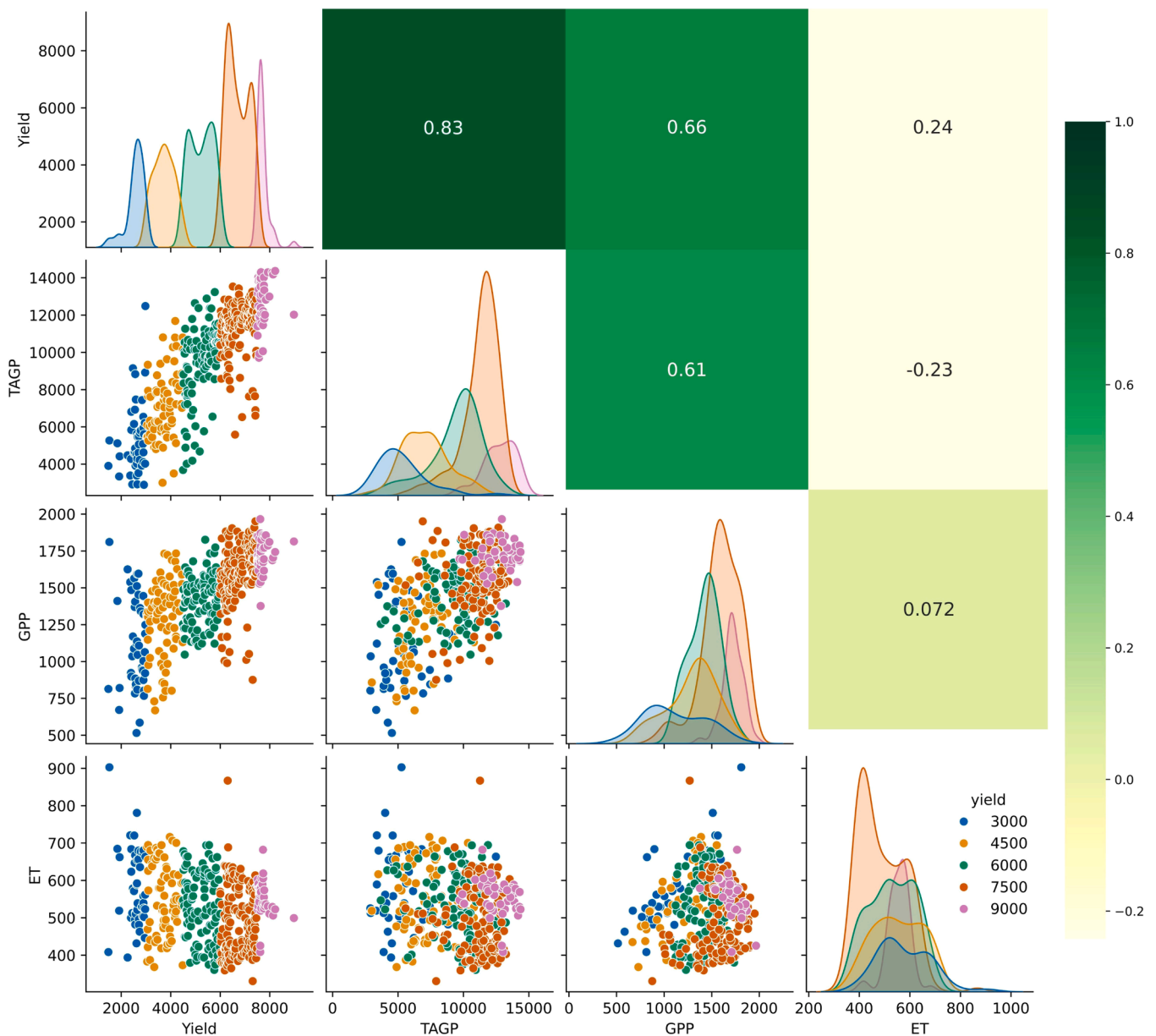


Fig. 4. Multivariate composite visual analysis diagram, including scatter plot matrix, kernel density estimation (KDE) diagram and Pearson correlation coefficient heat map.

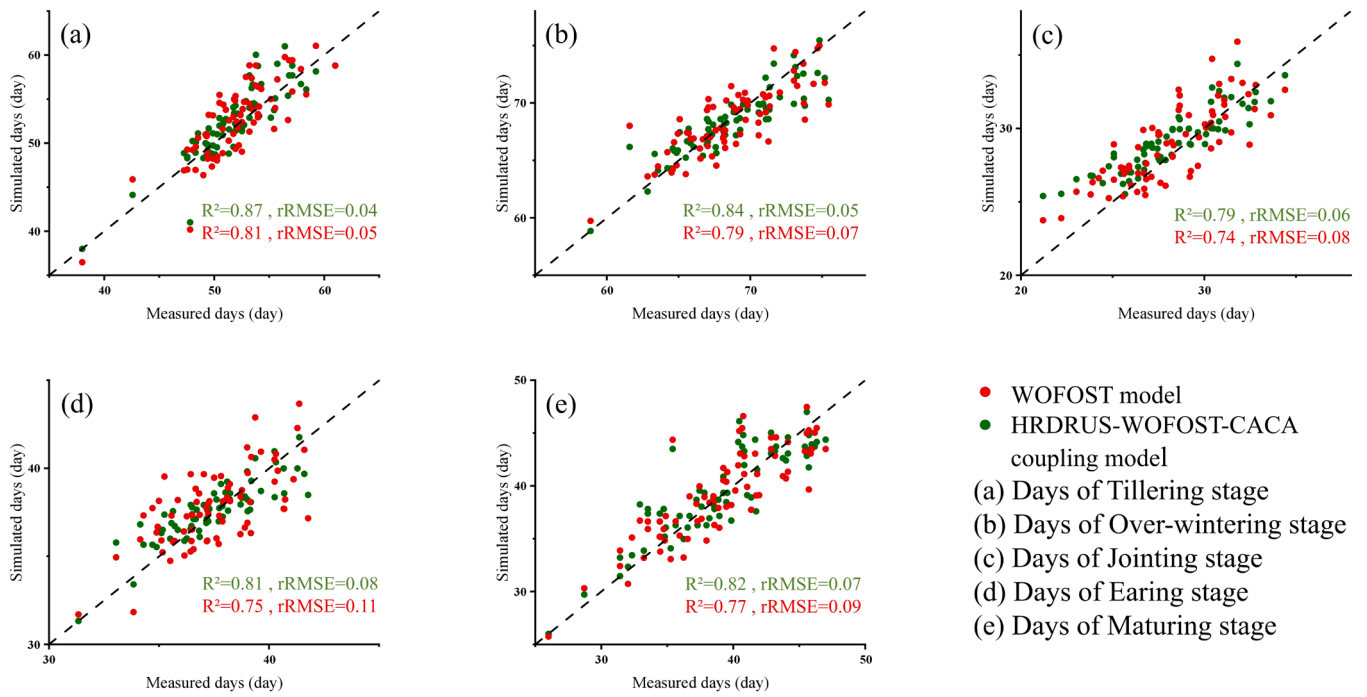


Fig. 5. Comparison of simulated growth period between WOFOST model and coupling model.

Fig. 6 illustrates the comparative performance of the WOFOST model, the coupled model, and the coupled model after data assimilation in simulating LAI and TAGP. Each subgraph depicts a distinct model and indicator. The green points in the figure represent the simulated values, with the abscissa representing the observed value and the ordinate representing the simulated value. The dotted line represents a 1:1

line, indicating that the simulated values are perfectly aligned with the observed values under ideal conditions. The figure demonstrates that the coupled model after data assimilation exhibits the greatest accuracy in simulating LAI and TAGP, followed by the coupled model. In contrast, the simulation accuracy of the WOFOST model is relatively low. The coupled model after data assimilation demonstrates superior

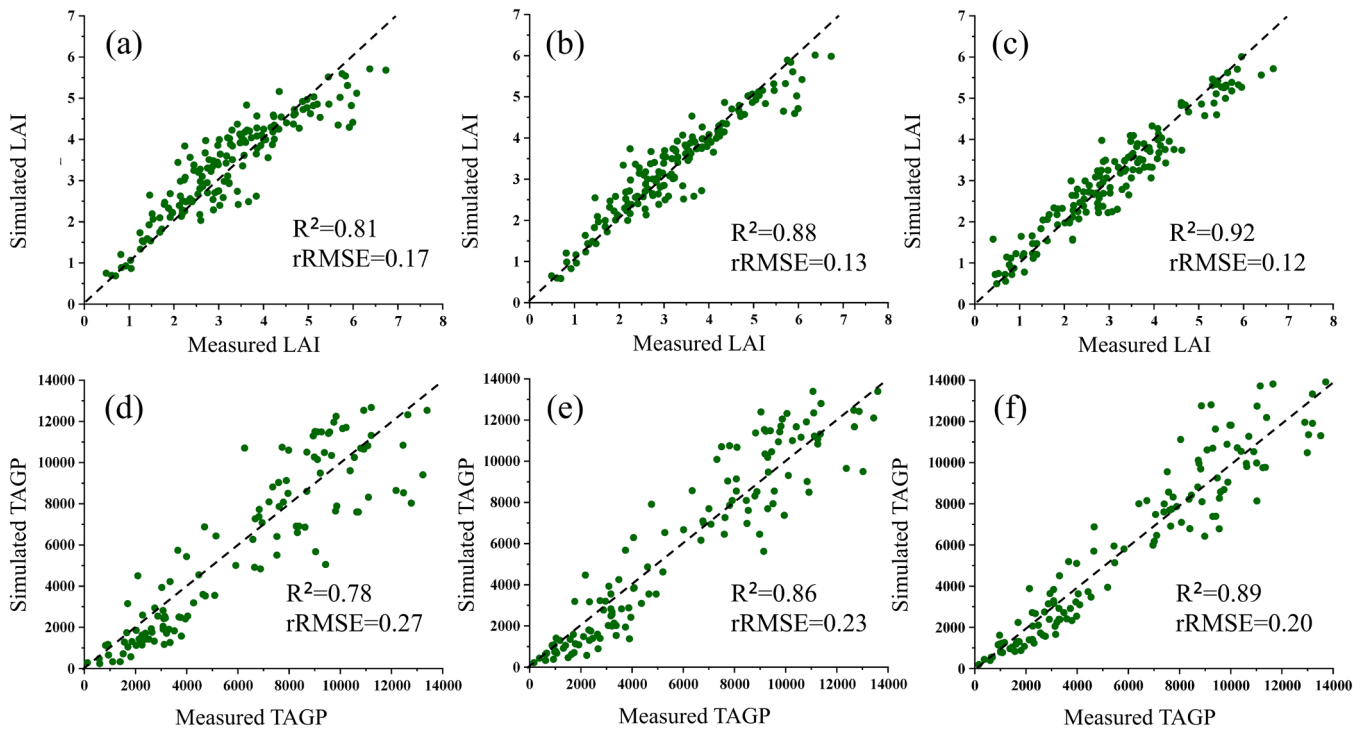


Fig. 6. Comparison of simulated LAI and TAGP between WOFOST model and coupled model and coupled model after data assimilation (a) Comparison of simulated LAI and observed LAI between WOFOST model (b) Comparison of simulated LAI and observed LAI between coupled model (c) Comparison of simulated LAI and observed LAI between coupled model after data assimilation (d) Comparison of simulated TAGP and observed TAGP between WOFOST model (e) Comparison of simulated TAGP and observed TAGP between coupled model (f) Comparison of simulated TAGP and observed TAGP between coupled model after data assimilation.

performance in terms of R^2 and rRMSE for each index, indicating a notable enhancement in simulation accuracy.

4.3. Data assimilation and parameter calibration effect verification

The Huyi District of Shaanxi Province is situated in the central and southern regions of the Guanzhong Plain, approximately 60 km from China's geographical centre. The climate is characterised by cold, warm and dry seasons, with abundant light, heat and water resources. This is representative of the production conditions typical of China's main wheat-producing areas (Yue et al., 2017). A statistical analysis of the mean temperature during the winter wheat filling period in Huyi District over the past three decades indicates that 2015 was the most closely aligned with the mean. Accordingly, this section employs the data from the agricultural station in 2015 to illustrate the impact of LAI assimilation on multi-source data during the growth of winter wheat. In comparison to the model that did not undergo assimilation, the LAI value exhibited a notable increase following the initial assimilation, rising from approximately 0.8 to 1.2. During the period of vegetation growth, the LAI simulation values following the second and third assimilations remained between 2.3 and 3.0, exhibiting minimal variation. However, during the fourth and fifth assimilations, a notable decline in LAI values was observed. This can be attributed to a discrepancy between the remote sensing LAI and the model simulation value, which decreased from approximately 4.0 to 3.2 after the fifth assimilation.

In general, the LAI value simulated by the model exhibits a high degree of correlation with the remote sensing data, particularly during the initial peak and subsequent stages. However, a discrepancy of approximately 0.5 remains between the two. This demonstrates that the model is capable of accurately reflecting the change trend of winter wheat LAI. Figure S2 illustrate the impact of LAI assimilation on total plant weight (WST), grain weight (WSO) and TAGP, and demonstrate the comprehensive adjustment outcome of the model subsequent to LAI assimilation. The maximum wheat aboveground biomass measured at the agricultural station was 13,255 kg/ha, while the simulated yield of non-assimilated wheat was 14,453 kg/ha. The absolute error was 1198 kg/ha, and the relative error was 9.03%. The yield of assimilated wheat was simulated to be 13,824 kg/ha, with an absolute error of 569 kg/ha and a relative error of 4.29%. The findings demonstrate that the predictive precision of the model for wheat growth can be markedly enhanced through the continuous assimilation of remote sensing data, thereby facilitating a more precise representation of the actual growth status.

The box plot presented in Fig. S3 illustrates the stability of the data at different growth stages, both before and after the assimilation process. To illustrate, during the wintering period, the median values before and after assimilation were approximately 0.1, and the box ranges (from -0.5 to 0.5) of the two were almost identical. The median and box range of the majority of stages demonstrate minimal variation, indicating that the assimilation process provides a robust foundation for data consistency.

A comparison of the connection lines reveals that certain data points, both preceding and following the assimilation process, have undergone modification. At the heading stage, a data point that had been adjusted from -1.0 to -1.5 before and after assimilation indicated that the latter process may have widened the gap between the data point and the overall trend. Following the process of data assimilation, the range of distribution of the TAGP data at varying growth stages was observed to undergo a slight expansion. To illustrate, the data points prior to assimilation were concentrated between -0.5 and 0.5, whereas the range subsequent to assimilation was expanded to -0.75 to 0.5. This reflects the model's enhanced flexibility in capturing dynamic changes.

Prior to the incorporation of remote sensing data into the output of the crop growth model, the primary factor influencing spatial variability is meteorological data. The calibration results serve to enhance the spatial variability within the meteorological grid while simultaneously

preserving the spatial trend. In comparison with the county statistical results, the results obtained prior to data assimilation have demonstrated an aptitude for encapsulating the overall spatial trend. However, due to the coarse resolution of the meteorological data, the capacity to express spatial details is somewhat limited. The planting density of winter wheat in the western region (Gansu, Xinjiang, Sichuan) is relatively sparse. The proportion of the planting area to the total administrative area is less than 1%, which renders the pixel information on the 500 m resolution map somewhat indistinct. Furthermore, as the model calibration is primarily based on statistical data, there are notable discrepancies in the spatial patterns observed in certain areas compared to the county-level statistical data from corresponding years prior to parameter calibration. As illustrated in Fig. 7(a), (b) and (c), this discrepancy is diminished following the data assimilation process.

4.4. Contributions of different indicators, phenological stages and machine learning methods to wheat yield prediction

Fig. 8(a) employs R^2 and rRMSE as metrics to assess the influence of disparate input indicator combinations on the efficacy of the Bagging Regressor model. The R^2 value for the TAGP + ET + GPP combination was 0.83, and the rRMSE was low. The R^2 and rRMSE of the TAGP + ET combination were marginally inferior to those of the aforementioned combinations, yet still demonstrated satisfactory performance. The R^2 and rRMSE values of the TAGP + GPP and ET + GPP combinations exhibited a further decline. The R^2 values of the individual indicators (TAGP, ET, GPP) ranged from 0.38 to 0.50, with rRMSE values exceeding those of the other indicators.

Fig. 8(b) illustrates the relative importance of each input index in the model. The weight assigned to TAGP is 0.617, indicating that it contributes the most to the model prediction results. The weight of ET is 0.217, which is the second highest after TAGP. The weight of GPP is the lowest, at 0.173. Although the contribution is relatively minor, it nevertheless exerts a discernible influence on the model. The combination of TAGP, ET and GPP can markedly enhance the predictive capacity of the model, whereas the impact of utilising these indicators in isolation is considerably inferior to that of the combined approach. In particular, TAGP plays the most pivotal role in the model, exerting a greater influence than ET and GPP. It is therefore necessary to use a coupling model and remote sensing data for a fine simulation of TAGP. Increasing the diversity of input variables can effectively improve the fitting degree and prediction accuracy of a wheat yield prediction model.

The R^2 of the Bagging Regressor was maintained above 0.75 up to 0.83, particularly on the 56th and 32nd days, with R^2 values exceeding 0.8, indicating high and stable prediction accuracy (Fig. 9). Concurrently, the rRMSE value exhibits fluctuations between 0.1 and 0.2, with a minimum value of 0.12. The rRMSE values on the 56th, 32nd and 24th days are lower, indicating a small and stable overall prediction error. The R^2 value of the decision tree ranged from 0.3 to 0.8, with a maximum value of 0.76. The R^2 values for the 56th, 40th, 32nd, 24th and 8th days exceeded 0.7, indicating a high degree of predictive accuracy. However, its rRMSE value exhibits considerable variation, ranging from 0.15 to 0.25, with a maximum of 0.24 and a minimum of 0.15. The rRMSE values on the 56th and 16th days are relatively low. Although the overall prediction error is minimal, the prediction performance exhibits considerable fluctuations. The R^2 values of Extremely Randomized Trees ranged between 0.3 and 0.7, with a maximum of 0.7. The R^2 values on the 32nd, 8th and 0th days were all greater than 0.6, indicating a relatively general level of prediction accuracy. The rRMSE value exhibits considerable variability, ranging between 0.15 and 0.25, with a maximum of 0.25 and a minimum of 0.17. Notably, the rRMSE on the 32nd and 8th days is relatively low, indicating a minimal prediction error. However, the overall prediction performance displays pronounced fluctuations.

Winter wheat yield prediction utilizing the WOFOST model demonstrated regional accuracy with a RMSE range of 953–1,503 kg/ha

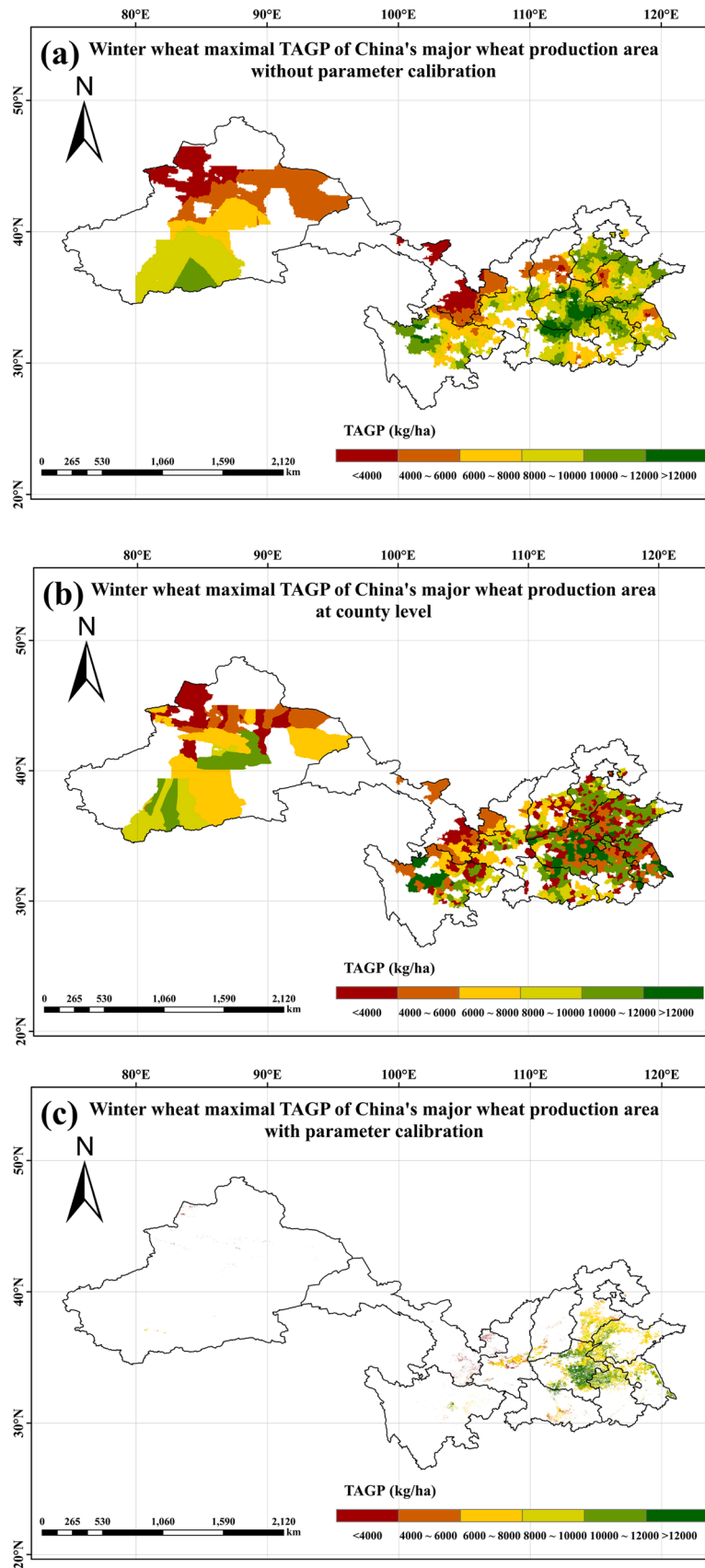


Fig. 7. (a) The results before data assimilation are generated by running the coupled model on the meteorological grid. (b) County-level results are generated by dividing the county-level statistical yield by the corresponding provincial harvest index. Note that some county data are missing. (c) The results of the assimilation data, the MODIS LAI is assimilated to the coupling model output in the grid, and the wheat planting area is used as the mask to cut the results.

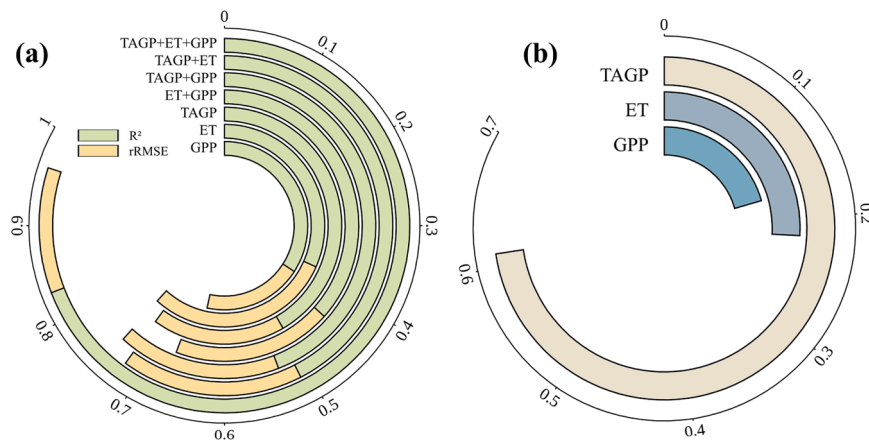


Fig. 8. (a) R^2 and rRMSE of different index combinations in Bagging Regressor regression prediction (b) Importance of each variable input to wheat yield estimation in Bagging Regressor regression prediction.

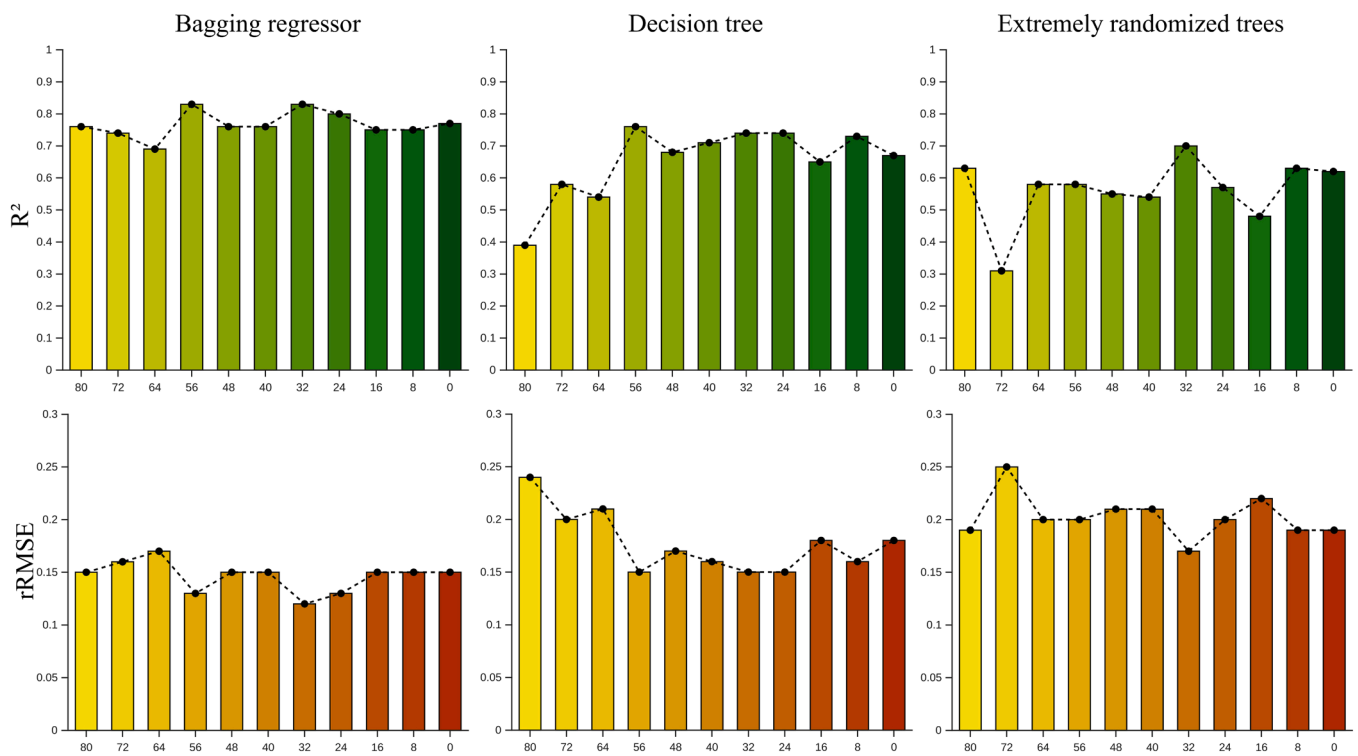


Fig. 9. The prediction effect of three machine learning methods on the 80 th to 0 th day before wheat harvest.

when validated against county-level statistical data spanning 2007–2015 (Huang et al., 2022). Validation of the modified CASA model for winter wheat yield estimation involved 29 sampling sites, yielding a coefficient of determination (R^2) of 0.56 and an RMSE of 1.22 t/ha (Wang et al., 2019). Notably, parameterization within the CASA model displays intrinsic variability under heterogeneous spatial scales, environmental regimes, and vegetation assemblages, as highlighted in prior research (Liu et al., 2017). The performance of GlobalWheatYield 4 km (Luo et al., 2022) was evaluated in comparison with the predicted yield of the yield estimation model proposed in this study. It can be concluded that, in comparison with the GlobalWheatYield4km model, the predicted yield of this model exhibited enhanced consistency and stability with the observed yield on the field scale (Fig. 10), and the yield prediction accuracy was relatively high (rRMSE = 0.17).

It is evident that there are fluctuations in the performance of each model on different days; however, they all demonstrate favourable

prediction accuracy on specific days (such as the 56th day and the 32nd day, corresponding to the heading stage and maturity stage of wheat, respectively). In general, the Bagging Regressor model demonstrates the greatest stability and accuracy in prediction. The performance of the Decision Tree and Extremely Randomized Trees models is satisfactory on certain days, but their overall prediction performance exhibits considerable fluctuations.

4.5. Uncertainty analysis in production forecasting

The application of all indicators (TAGP + ET + GPP) and the Bagging Regressor model at a prediction time of 32 days prior to maturity yielded a global Moran index of 0.301417 (Figure S4). The error is normally distributed, with a z-value of 78.81939, indicating that the probability of randomly generating this clustering pattern is less than 1%.

In general, the prediction model demonstrates superior capacity to

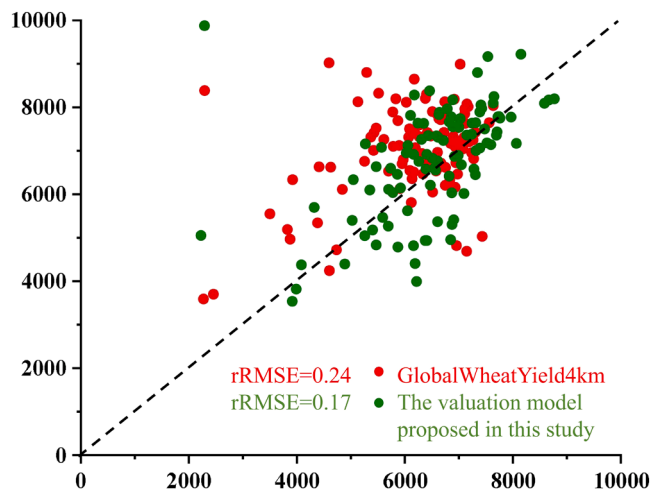


Fig. 10. The yield estimation model proposed in this study and the yield prediction accuracy of Global Wheat Yield 4 km.

capture the spatial heterogeneity of winter wheat yield. The discrepancy between the predicted and statistical yields at the county level is approximately between -10% and 15% (Fig. 11), with the majority of counties exhibiting an error of less than 1000 kg/ha. As illustrated in Fig. 12, the local Moran index map reveals the presence of notable high error areas (>15%) in southeastern Gansu, southern Shaanxi, eastern Sichuan, southwestern Anhui, and Hubei. The model prediction errors in these regions are substantial, which may be attributable to their distinctive climatic conditions. The predicted value of the model in the Huang-Huai-Hai region is lower than the actual yield, with an error range of between -10% and 0%. Conversely, the prediction effect in northern region is optimal.

5. Discussion

5.1. Uncertainty analysis of meteorological heterogeneity in yield prediction

It has been demonstrated that diverse climatic conditions exert a considerable influence on crop yield. Furthermore, the model's projection of wheat yield under various climatic types is also uncertain. As with the constraints of machine learning in elucidating the underlying physical processes (Jordan and Mitchell, 2015) the predictive inaccuracy of the model is considerable in regions characterised by pronounced

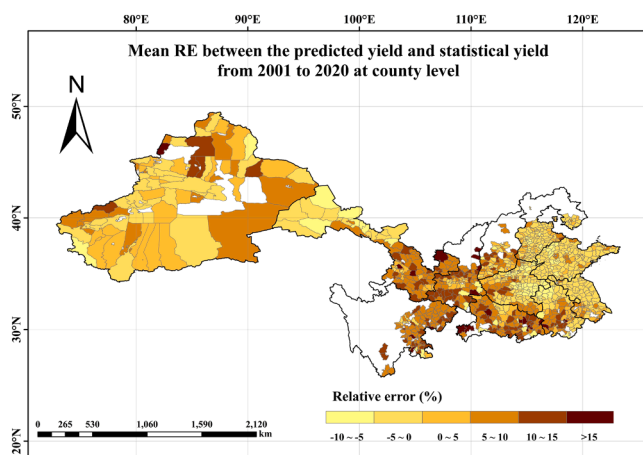


Fig. 11. Spatial distribution characteristics of average RE of predicted yield and statistical yield in main wheat producing areas of China from 2001 to 2020.

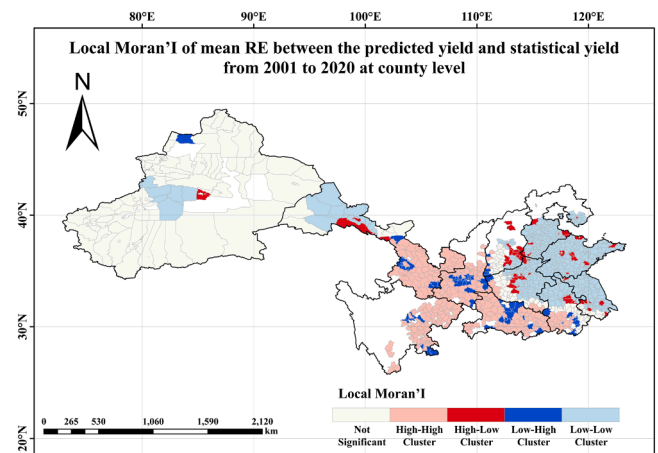


Fig. 12. The spatial distribution of local Moran's index of average predicted yield RE based on yield estimation model from 2001 to 2020 was analyzed.

climate heterogeneity, such as subtropical monsoon climates. This may be attributable to the climate variability in the region and the discrepancies in agricultural practices (Fan et al., 2022). For instance, the elevated degree of agricultural mechanisation and contemporary agronomic management techniques observed in the Huang-Huai-Hai Plain may result in the model underestimating the actual yield (Yue et al., 2022). It is acknowledged that traditional agricultural practices and the complex terrain in areas such as southeastern Gansu may serve to increase the prediction error of the model (Yang et al., 2024). As illustrated in Fig. 12, the model demonstrates a tendency to underestimate wheat yield in the temperate monsoon climate region, whereas it exhibits a tendency to overestimate in the subtropical monsoon climate region. In contrast, the model demonstrates a relatively high level of predictive efficacy in the temperate continental climate environment, which suggests that the climatic conditions in the region are well-suited to the model. These findings indicate that reliance on traditional climate and agricultural data may not provide an accurate representation of the actual situation in the region when predicting yield. To enhance the spatial adaptability and precision of the prediction model, future research should contemplate the integration of a broader array of environmental and physiological indicators into the model, including soil moisture and crop growth stage, which will facilitate a more accurate representation of regional characteristics and further optimise the prediction performance of the model.

5.2. The contribution of comprehensive climate variables to regional wheat yield prediction

Previous studies have described the impact of climate on vegetation through the use of climate data and vegetation indices (Phiri et al., 2020). However, these studies often fail to fully reveal the specific contribution of climate characteristics to regional differences. In contrast to the preceding analysis, which was based on a single climate index (Jolliet et al., 2018), this study employs a more comprehensive approach, encompassing the analysis of multiple climate variables. The principal components analysis revealed significant distributional differences in the meteorological data from the various provinces (Figure S1). The negative values of Gansu and Hebei on the PC1 axis indicate that their climatic characteristics are significantly different from those of other provinces. Conversely, the positive values of Henan, Hubei and Shandong indicate that the climate in these areas is similar. On the PC2 axis, the negative values of Gansu and Jiangsu are compared with the positive values of Sichuan and Xinjiang, thereby demonstrating the considerable impact of wind speed and precipitation on climatic characteristics (Zhang et al., 2021). The substantial loading of radiation on PC1 and PC2 suggests that it is a significant factor in the

interpretation of regional climate variations (Zanchettin, 2017). Accordingly, it is essential to establish the parameters of distinct geographical regions in accordance with the climatic conditions employed in the simulation of the coupled model. The present study revealed a significant positive correlation between wheat yield and TAGP ($R = 0.83$), a finding that aligns with those of previous research (Chen et al., 2021). While the TAGP index is effective in characterising crop growth, it may yield biased results when used in isolation. For instance, the yield estimation bias associated with the cumulative temperature, without threshold filtering may be observed (G. Poley and J. McDermid, 2020). As physiological indicators, GPP and ET demonstrate efficacy in yield prediction (Nguy-Robertson et al., 2015); however, their predictive accuracy is constrained by the uncertainty associated with the empirical formula parameters (Zhuo et al., 2022). In contrast, machine learning algorithms demonstrate greater resilience to these uncertainties and may therefore perform more effectively in yield prediction (Shahhosseini et al., 2019).

5.3. The advantages of the coupled model in dealing with environmental heterogeneity and its impact on crop prediction

The coupled model is an improvement on a single model in that it is capable of dealing with different environmental conditions. It has been demonstrated that the integrated utilisation of multiple models is an effective approach for addressing the heterogeneity of environmental and meteorological conditions (Bocquet et al., 2015). The HYDRUS model is an exemplar of soil moisture dynamics simulation, offering a detailed soil moisture movement and solute transport model that provides more accurate soil moisture data (Kandelous and Šimůnek, 2010). In comparison with the WOFOST model, HYDRUS is more effective at representing the spatial and temporal variations in soil moisture, which is essential for enhancing the precision of crop growth forecasting (Miller et al., 2019). The CASA model is capable of providing a comprehensive description of the processes of plant productivity, carbon fixation and soil respiration (Potter et al., 2012). The CASA model offers a more comprehensive simulation of net primary productivity and its response to environmental changes, as well as an enhanced assessment of ecosystem carbon dynamics (Liu et al., 2021). Nevertheless, the coupling model also introduces a certain degree of complexity, particularly with regard to parameter configuration and model interaction. It has been previously observed that the intricate nature of the model may result in an inherent unpredictability in the outcomes of the predictive analysis (Robinson et al., 2018). Accordingly, in order to enhance the efficacy and precision of the comprehensive model, it is essential to refine the parameter configuration and model interaction.

5.4. Data assimilation improves the prediction accuracy of winter wheat growth process

The model's prediction accuracy for the winter wheat growth process has been markedly enhanced through the continuous assimilation of remote sensing data. The LAI value simulated by the multiple assimilation method exhibits a high degree of correlation with the remote sensing data, particularly during the crucial growth phase (Huang et al., 2019). The principal factors contributing to this state of affairs are as follows: Firstly, the assimilation process effectively integrates remote sensing data and model simulation results, particularly during the crucial stages of winter wheat growth, such as the heading and filling stages (Wu et al., 2020). The model is therefore able to more accurately capture the dynamic changes of these key stages, thereby improving the simulation accuracy. Secondly, during the vegetative growth stage, the alteration in LAI is relatively minimal, thereby limiting the impact of assimilation on the precision of the simulation. However, during the reproductive growth stage, due to the considerable alterations in LAI, the assimilated model is capable of estimating LAI values with greater precision, thus enhancing the accuracy of yield prediction

(Xie et al., 2017). This finding aligns with the observation made in this study that the model's simulation accuracy of LAI demonstrates a gradual increase following multiple assimilations. Thirdly, although the assimilation of remote sensing data can enhance the precision of the simulation, it is also essential to consider the inherent uncertainty of the remote sensing data and the potential for error accumulation during the assimilation process. Such factors may have an impact on the modelling process and the accuracy of the predictions (Huang et al., 2019). It can be seen that there is no direct correlation between the length of the assimilation period and the accuracy of the predictions. In selecting an assimilation strategy, it is essential to consider both the accuracy of the predictions and the computational efficiency. In the future, we will attempt to couple machine learning surrogate models to enhance calibration efficiency, thereby fully leveraging the advantages of diverse remote sensing products.

5.5. The prediction effect of different machine learning models and multi-index combinations

This research realizes hybrid modeling of multimodal data via a dual bias-correction framework that synergizes mechanistic modeling and machine learning methodologies, thereby generating enhanced prediction stability. The Regressor algorithm is based on the bagging technology, which involves the construction of multiple parallel regression trees to estimate the model. The final prediction value is obtained by averaging the predictions made by all trees. This method is robust in dealing with limited samples and outliers, and reduces the variance of the model (Bakır et al., 2024). Although the decision tree performs well on some days, the overall prediction performance fluctuates considerably. This may be attributed to the fact that the single tree model is susceptible to fluctuations in the data, resulting in an unstable prediction (Basti et al., 2015). In contrast, Extremely Randomized Trees employs the integration of multiple trees, incorporating greater randomness into the selection of split nodes in each tree. This approach effectively enhances the diversity and prediction ability of the model, although it may also result in fluctuations in its performance (Saeed et al., 2021). The experimental results demonstrate that Bagging Regressor exhibits superior performance, effectively capturing the patterns in the data and providing more stable predictions. Prior studies have indicated that climate data and vegetation indices (such as NDVI or EVI) play a pivotal role in crop yield estimation (Shammi and Meng, 2021). In contrast, the physiological indicators employed in this study (such as GPP, ET and TAGP) are more directly pertinent to crop growth. Serving as physical constraint submodels, they ensure that the prediction results consistently remain within the feasible domain of biological processes. For example, evapotranspiration (ET) is calculated using the Penman-Monteith method, which combines meteorological data and vegetation information to enhance the precision of describing crop growth (Westerhoff, 2015). The combination of these indicators markedly enhances the predictive capability of the model, whereas the use of a single indicator is inadequate. This corroborates the benefits of multivariate combination in complex problem modelling (Tian et al., 2021).

6. Conclusions

In this study, a novel wheat yield estimation model was developed by integrating remote sensing data with physiological indices of wheat. The principal findings are as follows: Firstly, following data assimilation and parameter calibration, the HYDRUS-WOFOST-CASA coupling model demonstrated high simulation accuracy for wheat phenological period (average $R^2 = 0.83$, $rRMSE = 0.06$), LAI (average $R^2 = 0.92$, $rRMSE = 0.12$), and TAGP (average $R^2 = 0.89$, $rRMSE = 0.20$). This demonstrates that the model is capable of accurately capturing changes in wheat yield. Secondly, the integration of machine learning algorithms and multiple yield indicators, particularly in temperate continental climate zones, has

been shown to significantly reduce the error associated with yield prediction (average prediction error is 1.92 %). Furthermore, the model demonstrates robust spatial adaptability under diverse planting patterns and regional environments. Ultimately, the novel model demonstrates superior capacity to capture the spatial heterogeneity of wheat yield at the regional scale when compared to the GlobalWheatYield4km model. The findings demonstrate that the integration of crop growth models and remote sensing data-driven machine learning models enhances the precision of wheat yield forecasting.

The model established in this study provides effective technical support for wheat yield prediction in areas with a small-scale peasant economy and the coexistence of irrigation and rain-fed agriculture. It also demonstrates the timeliness, simplicity, and cost-effectiveness of regional wheat yield prediction using public data. This framework facilitates enhanced decision-making capacity in agricultural production and management, thereby providing guidance for food and water security. Further research could investigate the extent to which diseases and pests impact prediction, the influence of other remote sensing indicators, and the predictive capacity of alternative machine learning algorithms, thereby broadening the scope of applications for the findings of this study.

CRedit authorship contribution statement

Jiyuan Xie: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dongyan Zhang:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization. **Ning Jin:** Resources, Funding acquisition. **Tao Cheng:** Writing – review & editing. **Gang Zhao:** Writing – review & editing. **Dong Han:** Writing – review & editing. **Zhen Niu:** Writing – review & editing, Validation, Supervision. **Weifeng Li:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.agrformet.2025.110687](https://doi.org/10.1016/j.agrformet.2025.110687).

Data availability

Data will be made available on request.

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